Week 5 – Payment Transaction Management System

1. Payment Table Design

The Payment table stores payment transaction information such as Payment_ID, Order_ID, Payment_Date, Payment_Mode, Payment_Status and Transaction_Amount.

Payment_ID is the Primary Key, and Order_ID is a Foreign Key referencing the Orders table. The UNIQUE constraint on Order_ID ensures that each order has at most one payment record.

NOT NULL and CHECK constraints maintain data accuracy. Payment modes include UPI, Credit Card, Debit Card, Net Banking and Cash on Delivery. Payment statuses include Pending, Success and Failed.

2. Updated ER Diagram

ORDERS → PAYMENT

- Orders to Payment: One-to-One (optional payment record for each order)
- Order_ID connects the Orders and Payment tables.
- Each payment belongs to one order.
- An order can have zero or one payment record.

3. Sample Customer, Order & Payment Data

Sample customers include Arun Kumar, Priya Devi and Karthik Raj.

Sample payment transactions include:

- Order 1 – UPI – Success – ₹1,200
- Order 2 – Credit Card – Success – ₹9,000
- Order 3 – Debit Card – Failed – ₹1,500
- Order 4 – Cash on Delivery – Pending – ₹1,650

The payment records are linked to existing orders through Order_ID.

4. Payment Transaction & Revenue Analysis Reports

SQL queries are used to display successful payments, failed transactions and pending transactions.

Payment analysis includes the number of transactions by payment mode, the most preferred payment method, total successful payment revenue, revenue by payment method and average transaction amount.

Customer payment history displays customer details, order information, payment mode, payment status and transaction amount.

Additional reports count successful, failed and pending transactions and identify orders with failed or pending payments.

5. Query Output Screenshots

Screenshots from MySQL Workbench should be added for the following queries:

1. Successful Payments
2. Failed and Pending Transactions
3. Complete Payment Information
4. Transactions by Payment Mode
5. Most Preferred Payment Method
6. Successful Payment Revenue
7. Revenue by Payment Method
8. Average Transaction Amount
9. Customer Payment History
10. Successful, Failed and Pending Transaction Counts
11. Orders with Failed or Pending Payments